

BINARY INSTANCE-SPECIFIC COST LEARNING

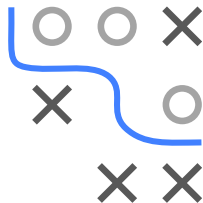
- Assumes instance-specific costs for every observation:
 $\mathcal{D}^{(n)} = \{(\mathbf{x}^{(i)}, \mathbf{c}^{(i)})\}_{i=1}^n$, where $(\mathbf{x}^{(i)}, \mathbf{c}^{(i)}) \in \mathbb{R}^p \times \mathbb{R}^2$.
- Define “true class” as cost minimal class
- Define observation weights: $|\mathbf{c}^{(i)}[1] - \mathbf{c}^{(i)}[0]|$

	$\mathbf{c}^{(i)}[0]$	$\mathbf{c}^{(i)}[1]$	$y^{(i)}$	$w^{(i)}$
$\mathbf{x}^{(1)}$	1	1	0	0
$\mathbf{x}^{(2)}$	1	2	0	1
$\mathbf{x}^{(3)}$	7	3	1	4

- Now solve weighted ERM:

$$\mathcal{R}_{emp}(\theta) = \sum_{i=1}^n w^{(i)} L\left(y^{(i)}, f(\mathbf{x}^{(i)} \mid \theta)\right)$$

- NB: Instances with equal costs are effectively ignored.



MULTICLASS COSTS

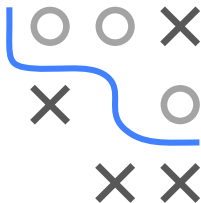
- Consider $g > 2$. Vanilla CSL is special case of instance specific, use $\mathbf{c}^{(i)}$ same for all $\mathbf{x}^{(i)}$ of the same class

		True class		
		$y = 1$	$y = 2$	$y = 3$
Pred. class	$\hat{y} = 1$	0	1	3
	$\hat{y} = 2$	1	0	1
	$\hat{y} = 3$	7	1	0

- For two $\mathbf{x}^{(i)}$ with $y = 2$ and $y = 3$:

	$\mathbf{c}^{(i)}[1]$	$\mathbf{c}^{(i)}[2]$	$\mathbf{c}^{(i)}[3]$	$y^{(i)}$
$\mathbf{x}^{(1)}$	1	0	1	2
$\mathbf{x}^{(2)}$	3	1	0	3
$\mathbf{x}^{(3)}$	1	0	1	2

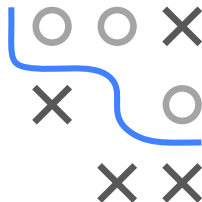
- Set $\mathbf{c}^{(i)}[y^{(i)}] = 0$, i.e. zero-cost for correct prediction.



- Let $\mathcal{D}^{(n)} = \{(\mathbf{x}^{(i)}, \mathbf{c}^{(i)})\}_{i=1}^n, (\mathbf{x}^{(i)}, \mathbf{c}^{(i)}) \in \mathbb{R}^p \times \mathbb{R}^g$.
- Example:

	$\mathbf{c}^{(i)}[1]$	$\mathbf{c}^{(i)}[2]$	$\mathbf{c}^{(i)}[3]$
$\mathbf{x}^{(1)}$	0	2	3
$\mathbf{x}^{(2)}$	1	0	1
$\mathbf{x}^{(3)}$	2	0	3

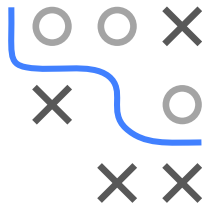
- Idea: Reduction principle to binary case (weighted fit) by one-versus-one (OVO).
- For class j vs. k :
 - How to deal with the label $y^{(i)}$? $y^{(i)}$ can be neither j nor k .
 - How to deal with the costs $\mathbf{c}^{(i)}[j]$ and $\mathbf{c}^{(i)}[k]$?



CISOVO

- When training a binary classifier $f^{(j,k)}$ for class j vs. k ,
 - Choose cost min class from pair $\arg \min_{l \in \{j,k\}} \mathbf{c}^{(i)}[l]$ as ground truth
 - Sample weight is simply diff between the 2 costs $|\mathbf{c}^{(i)}[j] - \mathbf{c}^{(i)}[k]|$
- Example continued:

	$\mathbf{c}^{(i)}[1]$	$\mathbf{c}^{(i)}[2]$	$\mathbf{c}^{(i)}[3]$	$\mathbf{c}^{(i)}[1 \text{ vs } 2]$	$\tilde{y}^{(i)}[1 \text{ vs } 2]$	$w^{(i)}[1 \text{ vs } 2]$
$\mathbf{x}^{(1)}$	0	2	3	0/2	1	2
$\mathbf{x}^{(2)}$	1	0	1	1/0	2	1
$\mathbf{x}^{(3)}$	2	0	3	2/0	2	2
	$\mathbf{c}^{(i)}[1]$	$\mathbf{c}^{(i)}[2]$	$\mathbf{c}^{(i)}[3]$	$\mathbf{c}^{(i)}[2 \text{ vs } 3]$	$\tilde{y}^{(i)}[2 \text{ vs } 3]$	$w^{(i)}[2 \text{ vs } 3]$
$\mathbf{x}^{(1)}$	0	2	3	2/3	2	1
$\mathbf{x}^{(2)}$	1	0	1	0/1	2	1
$\mathbf{x}^{(3)}$	2	0	3	0/3	2	3



- | | $\mathbf{c}^{(i)}[1]$ | $\mathbf{c}^{(i)}[2]$ | $\mathbf{c}^{(i)}[3]$ | $\mathbf{c}^{(i)}[1 \text{ vs } 3]$ | $\tilde{\mathbf{y}}^{(i)}[1 \text{ vs } 3]$ | $\mathbf{w}^{(i)}[1 \text{ vs } 3]$ |
|--------------------|-----------------------|-----------------------|-----------------------|-------------------------------------|---|-------------------------------------|
| $\mathbf{x}^{(1)}$ | 0 | 2 | 3 | 0/3 | 1 | 3 |
| $\mathbf{x}^{(2)}$ | 1 | 0 | 1 | -/- | - | 0 |
| $\mathbf{x}^{(3)}$ | 2 | 0 | 3 | 2/3 | 1 | 1 |

- ➊ For class j vs. k , transform all $(\mathbf{x}^{(i)}, \mathbf{c}^{(i)})$ to $(\mathbf{x}^{(i)}, \arg \min_{l \in \{j, k\}} \mathbf{c}^{(i)}[l])$ with sample-wise weight $|\mathbf{c}^{(i)}[j] - \mathbf{c}^{(i)}[k]|$.
- ➋ Train a weighted binary classifier $f^{(j,k)}$ using the above
- ➌ Repeat step 1 and 2 for different (j, k) .
- ➍ Predict using the votes from all $f^{(j,k)}$.

- $$\text{test costs of final classifier} \leq 2 \sum_{j \leq k} \text{test cost of } f^{(j,k)}.$$